

Methodology & Simulation

Section 3 speaking script

Deniz Ege Memetoğlu • FENG 498 • target time $\approx$ 3 min 45 s

Slide 1 • Section divider

$\approx$ 15 s

Calm hand-off, you take over from Deniz Etensel. You are now standing at the live model.

“Thank you, Deniz. We have seen the warehouse, the data, and the problem: the kits wait on the reach truck. My job was to turn that problem into something we can actually experiment on. So instead of moving a single real pallet, we rebuilt the whole kitting process as a discrete-event simulation in Python, using SimPy.”

Click to the flowchart.

Slide 2 • Flowchart: how one kit order is handled

$\approx$ 55 s

“This one diagram is the entire pipeline. A kit order arrives, and these orders do not come one at a time, they arrive in batches, exactly the way the real dispatch log shows. For every material in that kit, the model looks up where it is stored, and that location is decided by the slotting policy we are testing. Then one of three things happens. If the material is in the Kardex, an automated carousel brings it. If it sits low and within hand reach, an operator simply walks over and picks it. If it is up high, the operator has to wait for a reach truck. The operator assembles the full kit, walks it to the production line, and we record the KPIs: lead time, waiting time, walking distance, and utilization. Then the next order. Six policies, twenty independent runs each.”

Click to the SimPy logic slide.

Slide 3 • SimPy logic, distribution-driven

$\approx$ 80 s

“Now the part that makes this credible. The model is distribution-driven, it is not a replay. I did not record history and play it back. I fit statistical distributions from four months of the dispatch log, more than a hundred and sixty thousand rows, and I sample from them. That distinction is the whole reason this works: because the demand is generated and not fixed, I can move materials to new locations and re-run the same warehouse under a different policy, and the comparison stays fair.

A few numbers behind it. The arrivals follow the real batch pattern, and across the full twenty-run set the model lands at about four hundred orders a day, within roughly one percent of the real average of three hundred ninety-six. Pick times are not constant, they are lognormal, and the spread comes straight from a video time study of the F400 line, more than two thousand

timed micro-events, through this formula: $\sigma = \sqrt{\ln(1 + CV^2)}$. So the variability is measured, not invented. The model also holds the real fleet: eight operators, seven reach trucks, and four Kardex units, and any rack position can be picked by only one person at a time. And to keep the comparison airtight, every policy faces the exact same random demand, using common random numbers.”

Click to the live demo. The 3D scene starts playing at 0.25x.

Slide 4 • Live demo

≈45 s

Point at the screen on “follow one operator”, then pause 2–3 seconds and let them watch.

“And this is not a picture of the model, this is the model. What you are watching is the proposed policy running live, from straight overhead. Follow one operator. The picks stay low, on the manual shelves, right next to the line, and the reach truck is almost never called. That absence, the truck that does not move, is the entire result. Every number in the sections after me comes out of this engine.

With the method and the model in place, I will hand over to Nehir, who will show how each slotting policy is actually designed.”

Hand off to Nehir.

Pacing & notes

- About 470 spoken words, roughly 3 min 30 s to 3 min 45 s at a calm pace.
- If you run long, trim the fleet list in Slide 3 (“eight operators, seven reach trucks, four Kardex units”) down to “the real operator, reach-truck and Kardex fleet” to save about 15–20 seconds.
- **Slide check:** the slide currently prints “about 419 per day vs 396”. 419 is the analytic batch estimate; the measured full-run figure is about 400, within 1.1%. The script says “about four hundred, within one percent” on purpose, it is the more defensible number. Consider changing the slide bullet to “about 400 per day vs 396 real” so the screen and your words agree.

If the jury asks (sim part)

- *Replay or distribution-driven?* “Distribution-driven. I sample from fitted distributions, which is exactly what lets me re-slot and re-run counterfactually.”
- *Chi-square is significant, is the model wrong?* “No. It is significant because the model uses its own routing, not a bin-for-bin replay. Cramer’s V is 0.22, small to moderate. It is an internal-consistency check.”
- *RT utilization 1.3% seems too low.* “That is picking workload only. Replenishment is excluded equally for all six policies, so the comparison is fair; live total RT load would be higher.”
- *F400 timing extrapolated to nine families?* “Bounded by a plus-minus twenty percent sensitivity sweep. Line-aware stays first under every swing.”